

Scalability & Future Plan

Date	15 July 2024
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application currently supports only text-based emotion detection.	Users cannot detect emotions from speech or facial expressions.	High
2	The system relies on an active internet connection for Google Gemini API responses.	AI-generated guidance is unavailable when offline.	Medium
3	The application is designed for local deployment and has limited support for multiple concurrent users.	Scalability is limited for large-scale institutional use.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports single or limited users on a local machine.	Deploy on cloud infrastructure with support for multiple concurrent users.
2	Data Storage	Uses SQLite database and CSV backup files.	Upgrade to PostgreSQL or MySQL with cloud-based storage.
3	Performance	Models run on CPU with local resources.	Optimize models, enable GPU acceleration, and use model caching for faster inference.
4	Security	Basic API key protection using .env configuration.	Implement user authentication, encrypted data storage, HTTPS, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Implement cloud deployment and multi-user access using Streamlit Cloud or AWS.	3 Months	Makes the application accessible online for multiple users.
Phase 2	Add voice and facial emotion detection using speech and image processing.	6 months	Improves emotion detection accuracy and enhances user interaction.
Phase 3	Introduce multilingual support and personalized student profiles.	9 months	Expands usability for diverse users and delivers customized learning experiences.
Phase 4	Integrate advanced analytics, real-time notifications, and institutional dashboard.	12 months	Enables large-scale educational adoption with detailed insights and monitoring.